

PROBLEMS IN THE SELECTION AND UTILIZATION OF MATERIALS IN SODIUM COOLED FAST REACTORS

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1. Introduction

The concept of a sodium cooled fast reactor, fuelled with a mixture of uranium and plutonium oxides, led to the requirement for data on:

- (i) the behaviour of possible circuit and fuel cladding materials in dynamic liquid sodium in non-isothermal systems;
- (ii) the effect of neutron irradiation on the properties of potential cladding materials.

This paper describes the data which have been obtained on the compatibility of materials with sodium and also reviews the work which has been performed on the mechanical properties of cladding materials. The paper does not include data on the behaviour of fuel elements; this aspect, including the interaction between fuel and cladding, is covered by Lawton *et al.*,⁽¹⁾ this Symposium. Initially the designers required data on circuit materials at temperatures up to 560°C, and on cladding materials up to 650°C. We have obtained data on circuit materials at temperatures up to 650°C, and on cladding materials up to 725°C, in order to demonstrate the development potential of the system.

2. Choice of Materials

The initial choice of materials for use in fast reactors is influenced by the usual factors which are applied to any plant. Such factors are the existing information on the required physical and mechanical properties, thermal stability, fabrication experience, commercial availability and cost. Ultimately, however, the final choice of materials must depend upon the provision of

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good evidence that they will have satisfactory behaviour in sodium at the required operating conditions. In the case of core and cladding materials, the initial choice is also influenced by neutron economy; the final choice rests upon the provision of good evidence that irradiation to the peak neutron dose, which the materials will experience in service, will not so alter the original properties as to invalidate the bases of the original selection. Satisfactory compatibility of the cladding with the fuel under the relevant irradiation conditions must also be demonstrated.

TABLE 1. LIST OF MATERIALS USED IN THE INVESTIGATIONS

Section	Material	Heat treatment	Grain size, ASTM no.
1. Circuit materials	16/12/2 SS (316)	1050°C $\frac{1}{2}$ hour	
	18/8/Ti SS (321)	1050°C $\frac{1}{2}$ hour	
	2½Cr1Mo	(i) 950°C 1 hour (ii) 950°C 1 hour + 690°C 1 hour (iii) (ii) + 30% R. of A.	
	2½Cr1Mo0.4Nb 0.4 Ti	(i) 950°C 1 hour (ii) 920°C 1 hour + 690°C 1 hour (iii) 1150°C 1 hour + 650°C 1 hour	
2. Control rod materials	9Cr1MoV	(i) 800°C 1 hour (ii) 950°C 1 hour + 650°C 1 hour	
	Tantalum	1250°C 2 hours	
	Ta8W2Hf	1250°C 1 hour	
	Boron carbide		
3. Cladding materials (cf. Table 2)	316L	(i) 1050°C $\frac{1}{2}$ hour	5
		(ii) (i) + 17% R. of A.	5
	316	(i) 1050°C $\frac{1}{2}$ hour	6
		(ii) (i) + 17–20% R. of A.	6
	FV548	(i) 1075°C $\frac{1}{2}$ hour + 850°C 3 hours	9
		(ii) 58% R. of A. + 950°C $\frac{1}{4}$ hour	10
	Nimonic PE.16	1080°C 4 hours + 700°C 16 hours	6
	Nimonic 80A	(i) 1100°C $\frac{1}{4}$ hour (ii) 1040°C $\frac{1}{2}$ hour + 800°C 2 hours + 700°C 16 hours	— —

Circuit Materials

Since early work had shown that ferritic steels tend to lose their strength due to transfer of carbon to the surrounding sodium, even at temperatures as low as 400°C, most of our work on the compatibility of circuit materials with sodium has been confined to austenitic stainless steels and to ferritic steels containing at least 2½% chromium. The circuit materials, together with the control rod and cladding materials, which have been investigated are given in Table 1.

Cladding Materials

The object of the work on fuel cladding materials has been to develop a material which had properties such that the fuel element, which will be sealed, will maintain its integrity with little dimensional change throughout its life-time in the reactor. The initial target maximum burnup in an oxide fuel is 10% of the heavy atoms, achieved in about 10,000 hours, during which time the cladding will receive a total neutron dose of about 3×10^{23} n cm⁻² at temperatures between 370°C and 650°C. Clearly, the life of the fuel element

TABLE 2. TYPICAL CHEMICAL COMPOSITIONS OF CLADDING MATERIALS (wt. %)

	316L	316	FV548	Nimonic PE.16	Nimonic 80A
Cr	16.7	17.3	16.25	16.55	19.4
Ni	13.2	13.8	11.6	43.1	Bal.
Mo	2.4	2.41	1.3	3.35	—
Mn	1.6	1.8	1.08	0.05	0.05
Si	0.56	0.62	0.45	0.15	0.41
S	0.01	0.005	0.025	—	—
C	0.023	0.038	0.075	0.07	0.06
N	0.023	0.029	0.047	—	—
As	0.02	<0.005	—	—	—
Sn	0.03	<0.01	—	—	—
Co	0.02	0.014	—	0.2	—
V	0.01	0.033	—	—	—
P	0.018	<0.010	0.007	—	—
B	0.0002	0.0008	0.0037	0.0025	0.002
Nb	—	—	0.85	—	—
Cu	—	—	—	0.05	0.03
Al	—	—	—	1.24	1.21
Ti	—	—	—	1.19	2.31
Fe	Bal.	Bal.	Bal.	Bal.	0.53

will be determined by the capacity of the cladding to resist, without rupture or substantial deformation, the imposed stresses. These arise from the fission gas pressure, from the swelling of the fuel due to accumulation of fission products, and from temperature gradients. Stresses in the cladding can be varied by appropriate modification of the void/fuel ratio, cladding thickness and diameter, and the fuel rating.

If economies are to be made in the void fraction and cladding thickness, or if scope is to be allowed for the development of larger diameter elements operating at high fuel ratings, it is evident that high yield stress and creep strength are vitally important properties of the cladding. At first sight, ductility appears to be comparatively unimportant; the fuel elements are so close together that little change in element diameter can be permitted without

an unacceptable limitation on coolant flow. In fact a reasonably high ductility is desirable; first to allow for the incidence of accelerated local deformation which might occur at hot spots, or as a result of local mechanical interaction between fuel and cladding; secondly, because ductility exhaustion might occur due to strain cycling.

We have chosen two distinct classes of material, each representing a different approach to the problem of maintaining cladding integrity, for detailed investigation. These are:

- (i) austenitic stainless steels based on the AISI type 316 composition range, characterized by rather low yield stress, intermediate creep strength and high ductility;
- (ii) nickel-base alloys of the precipitation hardening type. These alloys have very high creep strength and yield stress values which compensate for their characteristically lower ductility. Examples of such alloys which have been examined are Nimonic 80A and Nimonic PE.16.

Typical compositions of the cladding materials which have been tested are given in Table 2.

3. Corrosion of Materials in Sodium

When alloys are exposed to sodium, mass transfer of metal occurs either by solution or by oxidation processes. Mass transfer of carbon can also occur. The reactor designer needs to know the extent of these processes and their effect upon the mechanical properties of the materials which he proposes to use in his design. We have therefore measured the dependence of mass transfer rates upon the velocity, temperature and oxygen content of the sodium with the aid of stainless steel loops in which the liquid metal is circulated. The design and operation of these loops have been described previously.^(2, 3)

Corrosion of Austenitic and Ferritic Steels

When stainless steels are first exposed to sodium containing about 25–30 ppm of oxygen, small initial weight gains and either black or green surface oxide films are observed. Under static conditions these films can persist. Electron microprobe examination of sectioned 18/8/Ti wires exposed to static sodium containing 25 ppm oxygen for 6 months at 650°C shows an outer chromium rich layer associated with an oxide. There is depletion of chromium in the region of metal immediately adjacent to this layer, with consequent relative enrichment of the matrix with respect to iron and nickel. This is shown in Fig. 1. X-ray diffraction studies indicate that the oxide films are sodium chromite (NaCrO_2).

Continued exposure of the oxide film to flowing sodium appears to reduce it progressively and it eventually disappears. After an exposure period of up to about a month, weight losses occur and subsequently the rate of metal loss is constant, if a steady temperature is maintained. Under these conditions the specimens are either matt grey metallic or shiny metallic in appearance, depending upon the oxygen content of the sodium. Electron microprobe

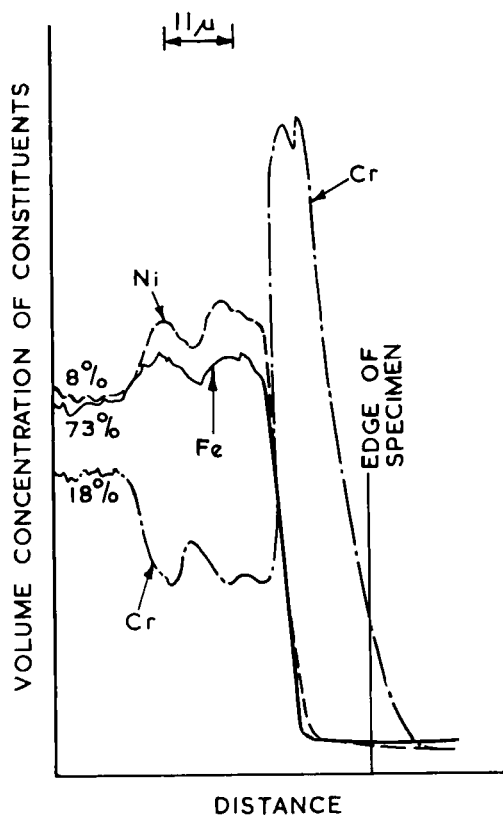


FIG. 1. Element depletion in the surface of 18/8/Ti stainless steel due to the formation of oxide film during exposure in static sodium containing 25 ppm oxygen at 650°C for 6 months.

examination of specimens of stainless steels, which have reached the steady state corrosion régime, shows that the surface layers are depleted in nickel and chromium and enriched with iron. Figure 2 shows the concentration gradients in the surface of two such specimens of type 316 stainless steel after exposure to sodium at 650°C; one specimen was exposed for 3 months in sodium containing 25 ppm of oxygen, the other was exposed for 6 months in sodium containing <5 ppm of oxygen. X-ray diffraction studies of the surface layers indicate that they are ferritic in structure and this is confirmed

by the fact that they are ferromagnetic. The growth of this ferritic layer with time is a matter which requires further study. It seems likely that the rate of growth will be determined by the parameters which control the diffusion of

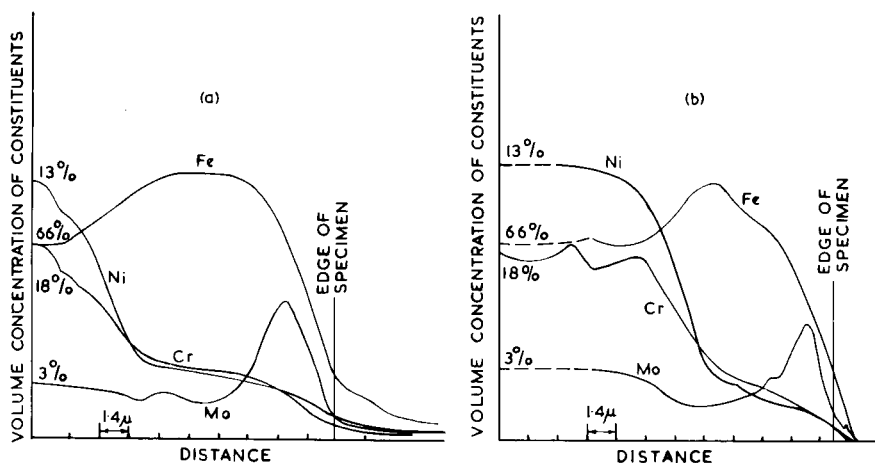


FIG. 2. Element depletion in the surface of type 316 stainless steel specimens after exposure to flowing liquid sodium at 650°C. (a) After exposure to sodium containing 25 ppm of oxygen for 3 months. (b) After exposure to sodium containing < 5 ppm of oxygen for 6 months.

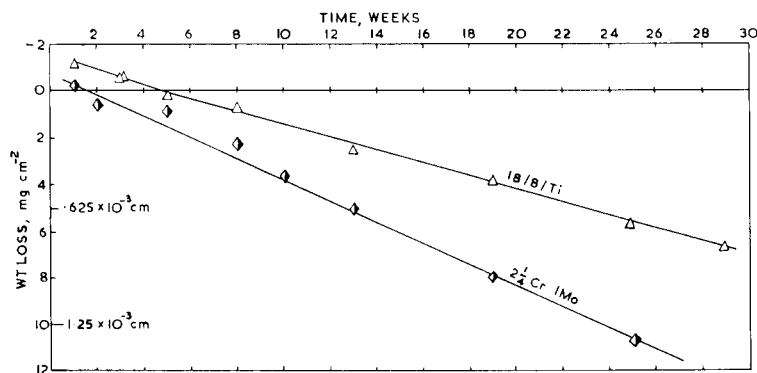


FIG. 3. The behaviour of 18/8/Ti stainless and $2\frac{1}{4}$ Cr 1Mo ferritic steels in flowing sodium at 600°C; O_2 concentration ~ 25 ppm; velocity 5.5 m sec^{-1} .

chromium and nickel through iron after an allowance has been made for the corrosion of the ferritic layer by the sodium.

$2\frac{1}{4}$ Cr 1Mo ferritic steel exhibits a constant rate of weight loss in dynamic sodium almost from the start of testing. On the other hand, black oxide films of sodium chromite form very readily on 9Cr 1MoV steel when it is

exposed to sodium containing 25–30 ppm oxygen. These films persist in flowing sodium for periods of up to 10 weeks before a constant rate of weight loss is observed.

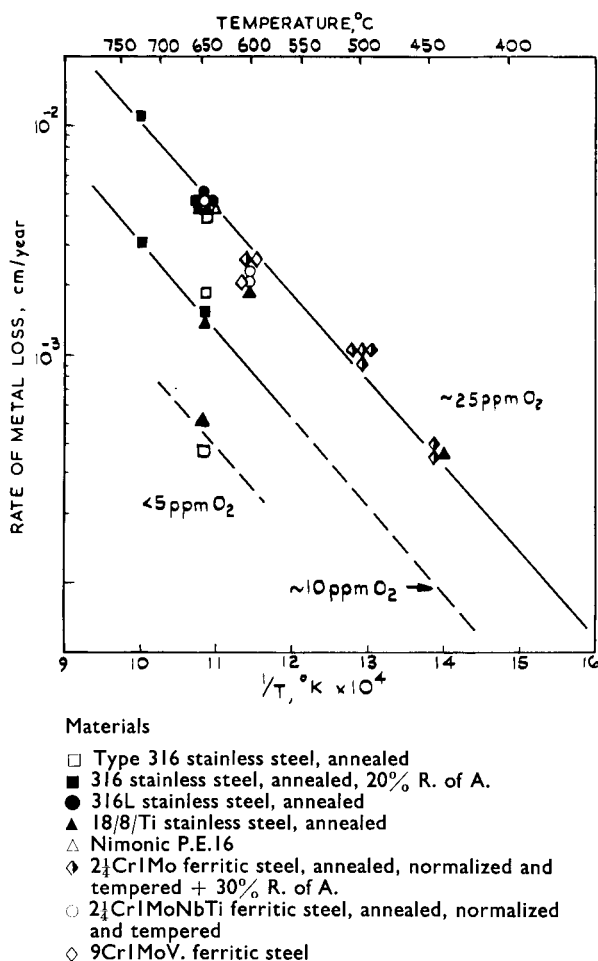


FIG. 4. The effect of temperature and oxygen concentration on the rate of metal loss from various materials after exposure to flowing sodium.

The corrosion data which we have obtained on both ferritic and stainless steels can be summarized as follows:

- (i) After an initial period, the corrosion rates of all the ferritic and stainless steels which we have tested are constant when steady conditions of sodium velocity, temperature, and oxygen concentration are maintained. This gives confidence in using a linear extrapolation to predict the long-term corrosion rates. Figure 3 illustrates the behaviour of

18/8/1 stainless steel and 2½Cr1Mo ferritic steel in tests lasting up to 6 months at 600°C.

- (ii) The steady state corrosion rates of all the ferritic and stainless steels, listed in Table 1, are very similar when the steels are tested under identical conditions. The metallurgical condition of each material does not affect the corrosion rate. These facts are illustrated in Fig. 4. This type of behaviour probably arises from the fact that the surface

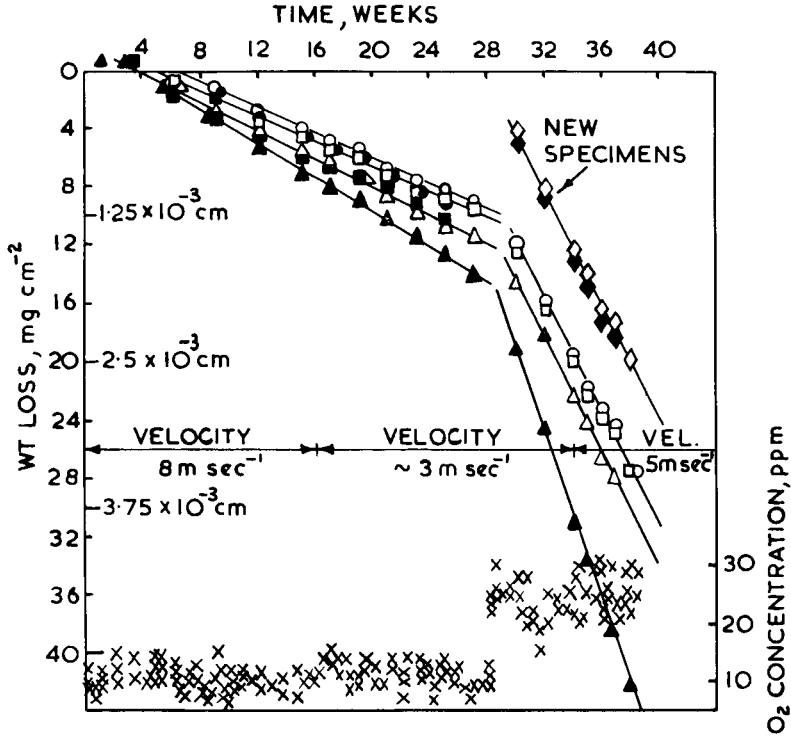


FIG. 5. The effect of changing the velocity and oxygen concentration of sodium on the rate of metal loss from 316 stainless steel; temperature 725°C; specimens *in situ* during the changes.

layers of all the materials are denuded of chromium and nickel and the steady state corrosion is then controlled, in all cases, by corrosion of the ferritic surface layer.

- (iii) The steady state corrosion rates are markedly dependent upon the oxygen content of the sodium. As shown in Fig. 4, in tests at 650°C, a change of oxygen content from 25 ppm to 10 ppm reduces the corrosion rate by a factor of about 3; a change from 25 ppm to less than 5 ppm reduces the rate by an order of magnitude. The effects are well exemplified in some tests done on 316 steel at 725°C where

the oxygen level was altered deliberately during the test from 10 to 25 ppm while keeping the temperature of the main circuit constant. It can be seen in Fig. 5 that a new and more severe corrosion rate was rapidly established.

- (iv) When the oxygen concentration and the velocity of the sodium are maintained constant, the steady state corrosion rates are very dependent upon temperature. Figure 3 shows that the corrosion rate, S , is related to the absolute temperature, T , in the temperature range 450–725°C, by the equation

$$S = 2.3 \times 10^{-6} (\exp) - \frac{17,500}{RT} \text{ cm sec}^{-1}; \quad (1)$$

in these tests the oxygen concentration was maintained at 20–30 ppm.

- (v) From experiments carried out at 650°C in sodium containing 10 ppm oxygen, the corrosion rate appears to be independent of sodium velocity when this exceeds 3 m sec⁻¹; below this velocity the corrosion rate is reduced and is dependent upon the velocity.

In our loops the isothermal test section is only 180 mm long. Corrosion rates in the specimens nearest the test section entrance are generally higher than those nearest the exit, but the variation in rate along the test section is very small. It is possible that corrosion rates in long lengths of pipework will fall below the values quoted above because the sodium may tend to become saturated with iron as it proceeds along the pipe. Down-stream effects of this kind have been reported by Lockhart and Billuris.⁽⁴⁾

Corrosion of Nickel-base Alloys

Nimonic PE.16 and Nimonic 80A have been tested in sodium at 650°C. As shown in Fig. 4, the corrosion rate of PE.16 in sodium containing 20–30 ppm oxygen, with a sodium velocity of 5 m sec⁻¹, is very similar to that observed in stainless steels. This is probably due to the fact that PE.16 contains about 33% of iron and that its corrosion rate is ultimately determined by the removal of a ferritic surface layer, after the surface has been denuded of nickel. This is a very tentative conclusion, and further work on the dependence of the corrosion rate of PE.16 upon temperature, sodium velocity, and oxygen content is required.

On the other hand, the corrosion rate of Nimonic 80A, which contains very little iron, is markedly different in character from that observed in stainless steels. In particular, the corrosion rate of Nimonic 80A is relatively insensitive to oxygen concentration and this is in line with observations by workers at the Fulmer Research Institute (private communication) that complex nickel sodium oxides are not stable in the presence of liquid sodium.

Nimonic 80A also differs from the other materials in showing a marked dependence of corrosion rate on sodium velocity, as illustrated in Table 3, so that for this material the corrosion process is apparently diffusion controlled. These results indicate that, at the sodium velocities likely to be required for fuel subassemblies, corrosion rates of Nimonic 80A will not be tolerable. No further results on this material will therefore be reported in this paper.

TABLE 3. THE EFFECT OF SODIUM VELOCITY AND OXYGEN CONCENTRATION ON THE CORROSION OF NIMONIC 80A AT 650°C

Velocity (m/sec)	Corrosion rate in sodium containing 10 ppm oxygen (cm/year)	Corrosion rate in sodium containing ~25 ppm oxygen (cm/year)
1.5	2.5×10^{-3}	2.5×10^{-3}
4.5	5.0×10^{-3}	—
7.5	6.9×10^{-3}	7.5×10^{-3}
12.0	12.7×10^{-3}	$13-15 \times 10^{-3}$

Corrosion of Control Rod Materials

Tantalum and boron carbide have been selected as the PFR control rod materials on the basis of their favourable neutron absorption properties. Some preliminary work has been performed on the behaviour of these materials in sodium containing 10 ppm of oxygen at 650°C. The corrosion rate of boron carbide in sodium flowing at 5 m sec⁻¹ is extremely high; a loss of 0.13 mm was observed after an exposure of 7 days. The metal loss in tantalum was 0.2 mm during an exposure of 1 month in sodium flowing at 7 m sec⁻¹ and 0.1 mm in the same period at 3 m sec⁻¹. A tantalum alloy containing 8% tungsten and 2% hafnium had better corrosion resistance than tantalum; the metal loss during 1 month's exposure to sodium flowing at 7 m sec⁻¹ was 0.1 mm. Clearly all of these materials will have to be protected by a very reliable cladding if they are to be used in sodium cooled reactors.

4. Transport of Carbon in Sodium

It has previously been stated by Harde and Stöhr⁽⁵⁾ that the use of low alloyed ferritic steels as circuit materials offers a number of advantages over the use of austenitic stainless steels. The major obstacle preventing the use of conventional ferritic steels, such as 2½Cr1Mo, is that they will transfer carbon to stainless steel via the surrounding sodium. If the high temperature parts of the circuit are constructed from stainless steel, transport of carbon from the ferritic steel is promoted by both the temperature gradients and the

chemical activity gradients. Therefore, in a system containing austenitic and ferritic steels, transport of carbon through the sodium leads to decarburization of ferritic steels, which contain carbides of high carbon activity; it also promotes the carburization of stainless steels and alloys containing high chromium contents, which contain carbides of lower carbon activity. Decarburization of the low alloyed steels and carburization of austenitic and high alloyed steels can have adverse effects on the mechanical properties of the materials.

Loss of strength in ferritic materials can be allowed for in detailed design. This solution is not completely satisfactory for mixed ferritic and austenitic circuits because the carbon released from the ferritic steel will carburize the austenitic steel. A more satisfactory solution appears to be the use of ferritic materials in which the carbon activity is comparable with that in stainless steel. Such steels are $2\frac{1}{4}\text{Cr1Mo}$ which has been stabilized by small additions of niobium and/or titanium, 9Cr1MoV steel and 12Cr1MoV steel.

Decarburization of Ferritic Steels

Several workers⁽⁶⁻⁸⁾ have previously reported decarburization of $2\frac{1}{4}\text{Cr1Mo}$ steel when it is exposed to sodium in stainless steel loops at temperatures above 500°C, and Gurinsky *et al.*⁽⁶⁾ reported that 5Cr0.5Mo steel decarburized at 560°C. We have observed marked decarburization of specimens of $2\frac{1}{4}\text{Cr1Mo}$ steel taken from stainless steel loops after an exposure of 26 weeks to sodium at 600°C; partial decarburization has been observed on specimens exposed at 450°C. The rate of decarburization does not appear to be very dependent upon the sodium velocity and oxygen concentration. In one experiment we have operated a loop containing pipes fabricated from $2\frac{1}{4}\text{Cr1Mo}$ and 18/8/Ti steels. The loop had three main temperature zones at about 520°C, 420°C, and 320°C and in each zone the surface area of ferritic steel was 1.7 times that of the austenitic steel. This loop was operated for 18 months and then some of the tubes were removed for examination. No decarburization was observed in the ferritic material at temperatures in the range 320–340°C. In the intermediate temperature zone at 420°C the ferritic steel pipes, which had an inner diameter of 17.5 mm and a wall thickness of 1.6 mm, had an average carbon concentration of 0.08% and a surface carbon concentration of 0.05% compared with the starting concentration of 0.12%. This demonstrates that substantial decarburization can occur in this material at 420°C.

Specimens of stabilized $2\frac{1}{4}\text{Cr1Mo0.4Nb0.4Ti}$ steel and 9Cr1Mo0.08V steel have been exposed to sodium in stainless steel loops; sodium velocities were 10 m sec⁻¹ and 0.15 m sec⁻¹ and the oxygen concentration was 20–30 ppm. No evidence of decarburization was observed in the stabilized $2\frac{1}{4}\text{Cr1Mo}$ steel either after 6 months' exposure at 600°C or in shorter term tests at 650°C.

The 9Cr1MoV steel was slightly carburized after 9 months' exposure at 600°C.

Carburization of Stainless Steels

When sodium is first introduced into our test loops, it contains a substantial amount of carbon. After the loops have been operated with about 1% of the flow passing through a cold trap for several days, the elemental carbon concentration in the sodium falls to <5 ppm. This carbon can cause carburization of the stainless steel specimens and pipework, and carbon which is removed from the sodium in this way is replaced by further carbon either from the cold trap or from other parts of the loop where carbon may have been deposited; thus the carbon concentration in the loops remains substantially constant if the cold trap temperature remains steady.

In our work on the carburization of stainless steels, small annular tensile specimens have been machined from bar material so that the specimen bore, through which the sodium flows, is cut from the centre of the bar. The carburization studies involve the determination, by chemical analysis, of carbon taken up from the sodium during an exposure period. Carbon concentrations are measured either as average bulk values or as average values in layers 0.01 cm thick machined from the exposed surfaces.

X-ray analysis has shown that the carbide near the specimen bore surface of 18/8/Ti material was of the $M_{23}C_6$ type; this was dispersed both within the grains and at grain boundaries at a higher density than in the body of the material. Examination of black deposits formed on type 316 stainless steel has shown that these have the composition Fe_4MO_2C . A detailed study of the parameters affecting the rate of carburization has not yet been performed, but present data indicate that the process is not very sensitive to sodium velocity or oxygen concentration.

The mechanical properties of materials which have been exposed to sodium for periods of up to 9 months at 650°C and 725°C have been compared with those of control materials which have had a similar thermal history in argon. The carburization has little effect on the UTS when this is measured at room temperature or 650°C. Loss of ductility in tensile tests at room temperature but not at 650°C has been observed and the results are summarized in Table 4. Data on Nimonic PE.16 are included in this table.

This carburization does not appear to cause any significant deterioration in the tube rupture properties of type 316 material. Sealed tubes containing high pressure argon, inner diameter 5 mm, wall thickness 0.375 mm, of type 316 material have been exposed to flowing sodium at 650°C for 6 months and their dilations are somewhat less than those observed in similar tubes which have been exposed to argon at 650°C for the same length of time.

It is not possible, at present, to say whether the combined effects of carburization and irradiation of fuel cladding will be serious. This aspect is

TABLE 4. CARBURIZATION OF STAINLESS STEELS AND NIMONIC PE.16 IN SODIUM AND ITS EFFECT UPON DUCTILITY

Material and original carbon level (wt. %)	Type of specimen	Sodium temperature (°C)	Oxygen level (ppm)	Time (months)	Carbon level after test (wt. %)		Room temperature elongation (%)			Elongation (%) at the sodium temperature		
					Mean	In a surface layer 0.1 mm thick	Original	In argon	In sodium	Original	In argon	In sodium
18/8/Ti 0.07	1 mm wall annular	650	< 5	6	0.09	0.14	59	57	39	—	—	—
		650	5	3	0.11	0.12	59	57	51	—	—	—
		650	10	4	0.12	0.18	59	57	52	—	—	—
		650	25/30	6	0.13	0.31	59	57	25	—	—	—
Annealed 316 0.05	1 mm wall annular	650	< 5	6	0.07	—	67	51	40	—	—	—
		650	5	3	0.08	0.10	67	51	53	—	—	—
		650	10	4	0.08	0.14	67	51	46	—	—	—
		650	25/30	6	0.09	0.19	67	51	33	—	—	—
Annealed 316L 0.01	1 mm wall annular	650	25/30	6	0.055	0.18	78	72	62	40	37.5	37.5
Annealed 316 0.05	0.37 mm wall annular	650	25/10	8	0.29	—	64	37	27	33	32	28
Annealed 316 0.05	0.37 mm wall annular	725	10/25	6½	0.32	—	52	40	31	36	25	23
				9	0.38	0.60	52	40	24			
				9½	0.38	—	52	40	24			
PE.16 0.05	1 mm wall annular	650	25/30	6	0.1	0.40	25/32	23	10	23/30	22	22

being investigated, but results are not yet available. It should be borne in mind, however, that the rate of carburization of fuel cladding in a reactor after several months of operation should be considerably less than that in our test loops. In the latter we maintained a relatively constant elemental carbon concentration of <5 ppm in the sodium. In a reactor the majority of the initial carbon can be removed from the sodium by cold traps in an auxiliary loop before the liquid metal is admitted to the main reactor circuits. The rate of carburization should increase with temperature; thus we would expect the carbon remaining in the sodium to be taken up mainly by the hottest parts of the circuit, namely, the fuel cladding. Therefore, although carburization may limit the life of the initial fuel charge, we expect that carburization of stainless steels will not be a significant long term problem in sodium cooled reactors, provided that the circuits do not contain materials which decarburize.

5. Mechanical Properties of Cladding Materials

Tensile Properties

The effect of long term annealing on the tensile properties of type 316 stainless steel and Nimonic PE.16 has been studied and the results are shown

TABLE 5. TENSILE PROPERTIES OF 316L AND NIMONIC PE.16 AFTER THERMAL EXPOSURE AT THE TESTING TEMPERATURE (STRAIN RATE = 1.58×10^{-3} sec $^{-1}$)

Ageing time (hours)	L. of P. (kg cm $^{-2} \times 10^{-3}$)		0.2% proof stress (kg cm $^{-2} \times 10^{-3}$)		UTS (kg cm $^{-2} \times 10^{-3}$)		Elongation on 1.6 cm (%)	
	650°C	700°C	650°C	700°C	650°C	700°C	650°C	700°C
Annealed 316L								
0	1.26	0.97	1.41	1.08	3.80	2.67	45	50
1000		1.23		1.27		2.81		57
2500	1.22		1.32		3.77		46	
4000	1.36	1.23	1.47	1.42	3.56	3.35	45	45
6000		1.34		1.43		3.30		42
10,000	1.35	1.29	1.45	1.45	3.53	3.44	41	37
17% pre-strained 316L								
0	3.11	2.92	3.20	3.09	4.11	3.66	31	34
1000	2.74	2.30	2.88	2.38	3.82	3.38	29	34
4000	2.65	2.25	2.72	2.32	3.84	3.47	30	34
6000	2.54	2.31	2.62	2.38	4.28	3.79	27	28
10,000	2.78	2.57	2.88	2.66	4.59	4.43	25	22
Nimonic PE.16								
0	4.34	4.41	4.41	4.54	7.61	7.18	33	39
1000	5.06	5.04	5.30	5.20	8.36	7.44	22	27
2500	5.20	5.67	5.52	6.05	8.40	7.80	21	27
5000	6.07	5.32	6.30	5.64	8.60	7.61	20	23
7500	—	—	5.65	5.20	8.44	7.52	21	26
10,000	—	—	5.41	4.92	7.87	6.92	28	32

in Table 5. Annealing of solution treated 316L material (see Table 1) at 650°C for periods of up to 10,000 hours has no detrimental effect on the tensile properties. After annealing at 700°C metallographic examination reveals precipitates of sigma and chi phases as well as the grain boundary

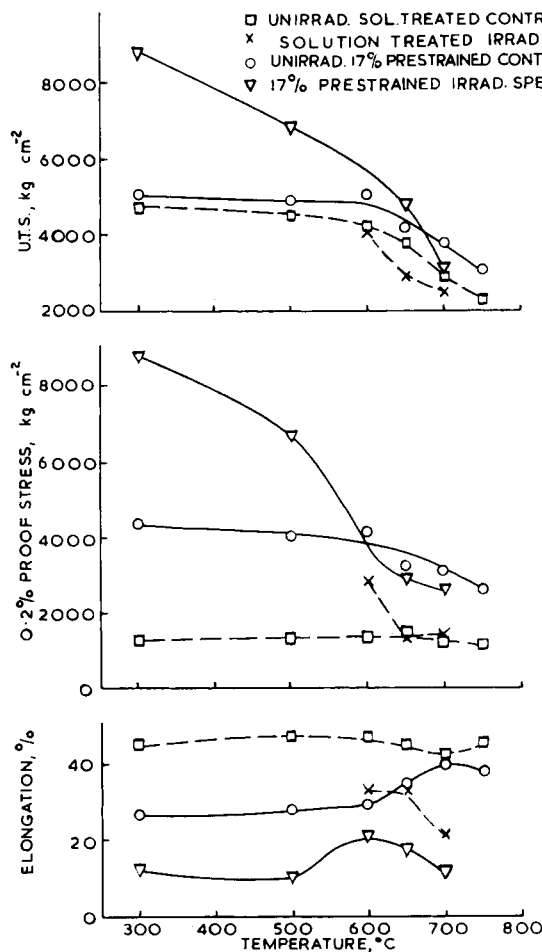


FIG. 6. The tensile properties of solution treated and 17% prestrained 316 steel after irradiation to a total dose of 3.6×10^{22} n cm⁻² in DFR (initial strain rate 1.58×10^{-3} sec⁻¹).

carbides which appear at 650°C. The presence of these phases increases the ultimate tensile strength and reduces the ductility. Soaking cold worked 316 material at 650°C produced only small changes in proof stress and ductility and no grain growth was observed; at 700°C some recrystallization occurred and there was a gradual deterioration in properties, up to 4000 hours, but

annealing for longer times recovered part of the loss in proof stress. Soaking Nimonic PE.16 increases the proof stress and decreases the ductility initially at both 650°C and 700°C, but after prolonged annealing the properties tend to revert to their original values; nevertheless, even after 10,000 hours the proof stresses are higher than the original values at both temperatures. Thus annealing at temperatures up to 700°C for periods of up to 10,000 hours is not likely to cause any marked deterioration in the tensile properties of the potential cladding materials which we are investigating.

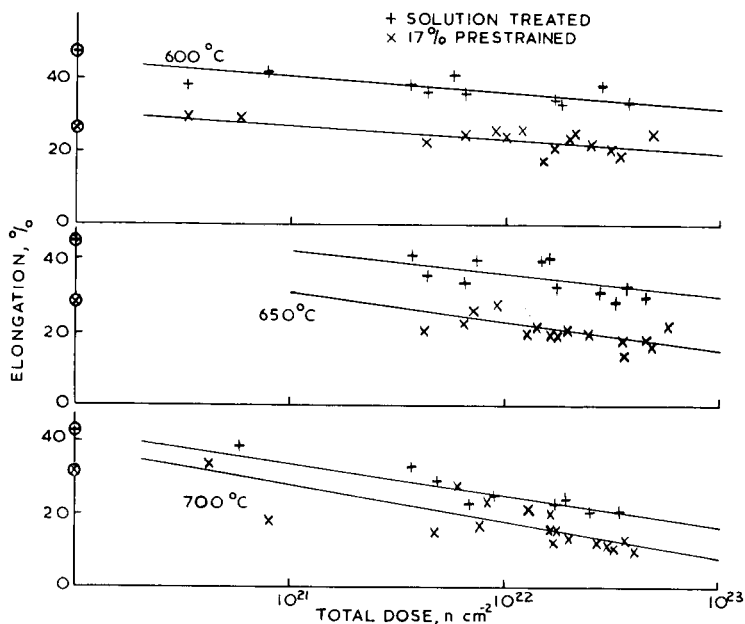


FIG. 7. The effect of neutron dose on the tensile elongation of solution treated and 17% prestrained 316 steel irradiated in DFR (initial strain rate $1.58 \times 10^{-3} \text{ sec}^{-1}$).

The effect of irradiation on the short term tensile properties of cladding materials has been investigated up to a maximum total neutron dose of $5 \times 10^{22} \text{ n cm}^{-2}$ in the Dounreay fast reactor (DFR). Most of the data have been obtained from materials irradiated at temperatures below 400°C, but some results have been obtained on specimens irradiated to a maximum dose of $1.6 \times 10^{22} \text{ n cm}^{-2}$ at about 600°C.

It can be seen from Fig. 6 that irradiation of type 316 stainless steels in DFR increases the proof stress and ultimate tensile stress (UTS) and reduces the ductility at test temperatures below 600°C. The change in low temperature mechanical properties appears to be a function of total dose but saturation of the hardening effect is evident at a dose between 2 and $3 \times 10^{22} \text{ n cm}^{-2}$, at which stage yield and UTS values are virtually identical. At test tempera-

tures above 600°C continued irradiation in DFR has little effect on the yield and UTS values but it does result in a progressive decrease in ductility. This reduction in elongation values becomes more marked as the test temperature

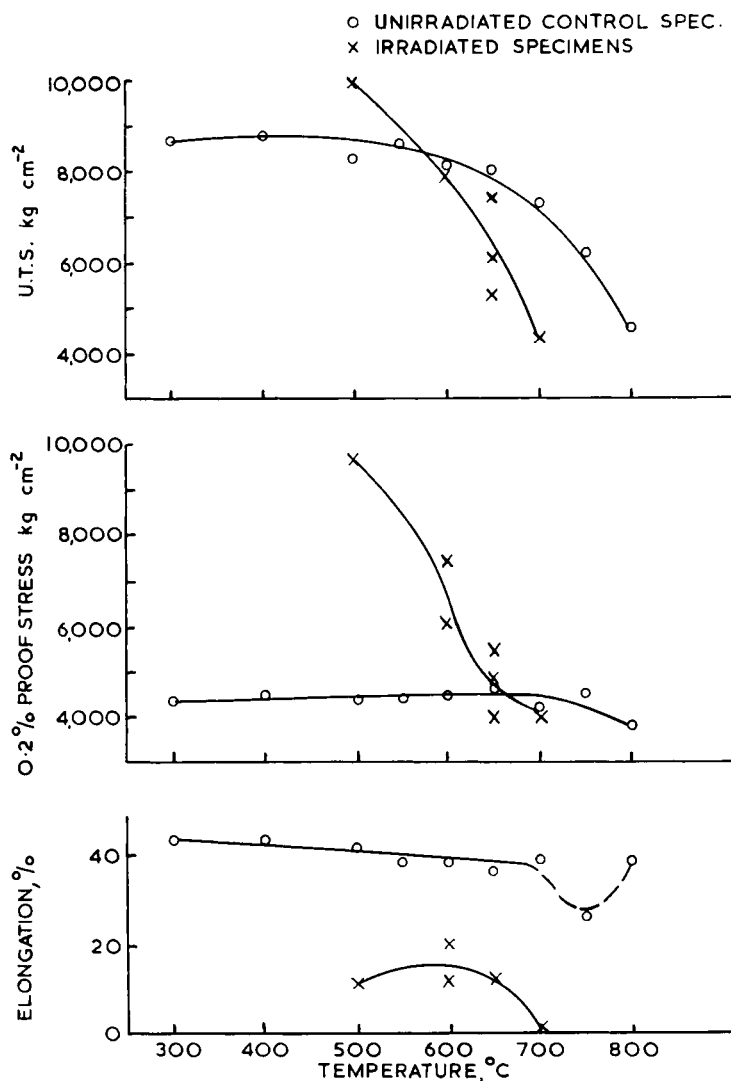


FIG. 8. The tensile properties of Nimonic PE.16 after irradiation to a total neutron dose of about $41 \times 10^{22} \text{ n cm}^{-2}$ in DFR (initial strain rate $1.58 \times 10^{-3} \text{ sec}^{-1}$).

is raised and, as shown in Fig. 7, appears to be a linear function of the logarithm of the total dose. The effect of irradiation on solution treated and cold worked materials appears to be generally similar, although at high

neutron doses and low test temperatures, the properties of the two types of material are indistinguishable. Tests on materials with small grain sizes strongly suggest that a decrease in grain size reduces the irradiation induced loss in high temperature ductility. Similarly, irradiation of FV548 in thermal reactors suggests that the presence of finely divided (200 Å) niobium carbide in the grain boundaries of this material makes it less susceptible than 316 material to the loss of high temperature ductility. In the same way precipitation of grain boundary carbide during irradiation at high temperature appears to be beneficial in retarding the loss in high temperature ductility.

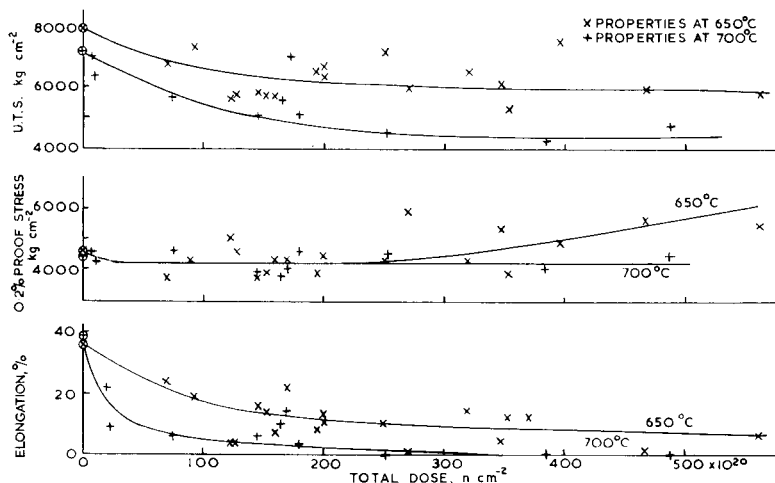


FIG. 9. The effect of neutron dose on the tensile properties of Nimonic PE.16 irradiated in DFR.

The effect of irradiation on Nimonic PE.16 at low temperature in DFR is shown in Fig. 8. Marked deterioration in properties is observed on testing at temperatures above 600°C and after a neutron dose of 4×10^{22} n cm⁻² the ductility at 700°C is reduced to negligible values, while the UTS at this temperature is about 60% of its original value. The degradation in properties increases progressively with neutron dose as shown in Fig. 9. Irradiation of this alloy at 650°C in thermal reactors and at 600°C in DFR indicates that the irradiation temperature is very important because precipitation of the hardening phase during irradiation appears to inhibit the processes which lead to loss of strength and ductility at high temperatures.

Creep and Rupture Properties

Extensive creep and rupture testing has been performed on the cladding materials listed in Table 1. Most of the work has been performed on unirradiated material. Examination of the individual creep curves shows that,

for most of the materials tested, particularly at stresses below 600 kg cm^{-2} , the primary creep strains are very small. This permits the use of secondary creep rates to calculate creep strains in fuel element cladding. The relationship between stress and secondary creep rate can be expressed in the form

$$\dot{\epsilon} = A\sigma^n, \quad (2)$$

where σ is the stress and A and n are constants at a given test temperature. Values of A and n for the various materials at different temperatures are given in Table 6.

TABLE 6. CREEP PROPERTIES OF CLADDING ALLOYS

Material	Condition	Temperature (°C)	($\text{h}^{-1} \text{ kg}^{4n}; \text{cm}^{2n}$)	n
316L	Solution treated	600	1.1×10^{-22}	5.5
		650	2.6×10^{-17}	4
		700	1.1×10^{-16}	4.2
		800	8.2×10^{-14}	3.8
316	Solution treated	650	2.4×10^{-23}	5.9
		700	1.9×10^{-22}	5.9
	20% R. of A.	650	2.4×10^{-41}	11
FV548	Solution treated and aged	650	2.3×10^{-22}	5
		700	2.6×10^{-21}	5
PE.16	Solution treated and aged	650	1.5×10^{-69}	18
		700	1.5×10^{-66}	18

In a sealed fuel element in a fast reactor the hoop stress due to build-up of the pressure of gaseous fission products will increase with time. It has been established for 316L material at 650°C that increasing the stress linearly with time does not provide repeated increments of primary creep. Therefore, since primary creep strains are very small at the stresses expected in a fast reactor fuel element, if the pressure increases with time at a known rate, the strains in a fuel element can be calculated by integrating equation (2). The rate of increase of pressure depends upon the detailed design of the fuel element; it can be shown that even if all the gaseous fission products are released from the fuel, creep strains in 316L should be less than about 0.3% at 650°C after 10,000 hours' operation at the peak flux position in a PFR fuel element, if irradiation does not affect the creep strength (cf. Bishop *et al.*⁽⁹⁾). The other materials will have even lower strains under the same circumstances.

A convenient way of comparing the creep strengths of various materials is shown in Table 7. This table gives the stresses, at 650°C and 700°C , required

to produce a strain rate of 10^{-6} h^{-1} in the different materials. It shows that the 316L material has the lowest creep strength. 316 material is better than the 316L material, presumably because of its higher carbon content. 316 material containing 20% cold work has a much higher creep strength than solution treated material at 650°C , but the improvement is less marked at 700°C : FV548, which is similar in composition to 316 with the addition of about 1% niobium, has a considerably higher creep strength. This should be further increased by cold working the material. FV548 has been studied in two grain sizes. Table 7 shows that a decrease in grain size gives a slight deterioration in the creep strength at 700°C . Table 7 also shows that the creep strength of the Nimonic PE.16 alloy is much higher than that of the stainless steels.

TABLE 7. STRESS TO PRODUCE A SECONDARY CREEP RATE OF 10^{-6} h^{-1} IN CLADDING MATERIALS AT 650°C AND 700°C

Material	Condition	Stress (kg cm^{-2}) for secondary creep rate of 10^{-6} h^{-1}	
		650°C	700°C
316L	Solution treated	430	250
316	Solution treated	560	420
	20% R. of A.	1400	—
FV548	Solution treated and aged (ASTM 9)	1300	850
FV548	Low temperature solution treated (ASTM 10)	~ 1500	~ 650
PE.16	Solution treated and aged	3000	2100

The rupture properties of tubes, inner diameter 5 mm, wall thickness 0.375 mm, have been studied by subjecting them to internal gas pressure at 650°C and 700°C . The ductilities of the tubes are much lower than those observed in uniaxial stress rupture tests. The presence of 20% cold work markedly improves the rupture life of 316 material, but the ductility of the cold worked material is only 2–3% compared with 10% in solution treated material. The rupture life data on the latter material show considerably more scatter than cold worked material and this appears to be due, at least partly, to local variations in wall thickness. FV548 tubes in the solution treated condition have comparable rupture strengths, in tests of 1000 hours' duration, to cold worked 316 tubes at 650°C , and the ductility, which is about 10%, is much higher. PE.16 tubes have the highest rupture strengths with a failure ductility of 5–12%. All the materials exhibit a change in mode of fracture as the stress changes. At high stresses failure is associated with considerable local grain deformation near the fracture. At lower stresses

failure is by the formation of a fine leak path associated with the formation of intergranular fissures. The stress at which this transition occurs is dependent upon the material and the temperature, and in general lies well above the stress level that is expected to arise from gas pressure in a PFR fuel element. Thus if failures do arise due to gas pressure, we expect that they will be of the kind which yield small leaks of fission product gases due to the formation of small intergranular fissures.

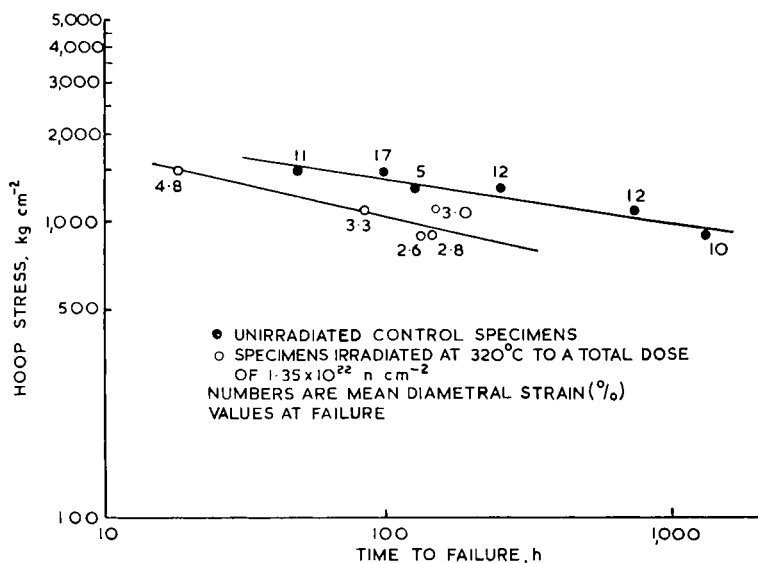


FIG. 10. The effect of irradiation on the rupture life of 316L tubes at 650°C.

The effect of fast neutron irradiation on the rupture properties of 316 material has been studied in post-irradiation rupture tests on bar specimens and tubes which have been irradiated to a total neutron dose of $1.3 \times 10^{22} \text{ n cm}^{-2}$ at temperatures below 400°C in DFR. Present data are confined to solution treated material. The irradiation reduced the rupture life of bar specimens by a factor of 2 to 3 with a similar loss in ductility. Results for 316L pressurized tubes are shown in Fig. 10. The irradiation reduced the rupture life at a given stress by a factor of about five and reduced the rupture ductility from 5–17% to 2–5%. Rupture ductilities were derived from careful measurements on the tube diameter at regions remote from those of high local strain at the point of failure. A number of tests were interrupted at intervals in order to measure the strain rate. These revealed that primary creep strains were very small, so that terminal strain could be used to calculate approximate creep rates. Present results suggest that the irradiation has enhanced the creep rate by a factor of between 2 and 3.

Obviously much more information of this type is required on the materials after they have been irradiated to higher doses at higher temperatures.

Some experiments have been performed on helical springs which were irradiated under tensile stress in DFR. The materials tested were 316 stainless steel, Nimonic PE.16, and pure molybdenum. Stresses of 350 kg cm^{-2} were applied to the springs, which experienced a total neutron dose of about $10^{22} \text{ n cm}^{-2}$. Interpretation of the results has been frustrated by uncertainties in the determination of the irradiation temperature. This was $\pm 25^\circ\text{C}$ and this implies that only a major effect could be detected. However, it may be inferred, from a comparison of the amount of creep undergone by different materials in the same rig, that the presence of a neutron flux during stressing does not have a major effect on the creep rate of stainless steel or PE.16. At worst, the effect due to irradiation at temperatures between 575°C and 650°C appears to be no greater than the effect which could be ascribed to the imprecision in the measurement of irradiation temperature.

6. Mechanisms of Radiation Damage in Cladding Materials

Two distinct types of effect are produced in austenitic steels and nickel base alloys when they are irradiated in a fast neutron spectrum. It is convenient to discuss these separately.

Low Temperature Effects

At temperatures below about 400°C neutron irradiation increases the strengths and reduces the ductilities in tensile tests which are performed after irradiation at absolute temperatures up to about half of the melting point. These property changes are caused by the formation of vacancies and interstitial atoms by collisions either between neutrons and lattice atoms, or between energetic displaced atoms and lattice atoms. The point defects migrate thermally to form clusters which impede dislocation movement, thereby promoting the strength increases and the reductions in ductility. The clusters can be resolved by electron microscopy using thin film techniques. The mechanism of the point defect-dislocation interactions in metals and alloys, which has already been described previously by Barnes⁽¹⁰⁾ will not be discussed in detail here.

This type of radiation damage can be removed progressively by annealing at temperatures in the range 400 – 700°C . During the annealing process dislocation loops are formed. These increase in size and decrease in number as the annealing temperature is raised.

High Temperature Effects

It is now well established that neutron irradiation in thermal reactors also reduces the tensile ductilities of austenitic steels and nickel base alloys at

temperatures above about 550°C.^(11, 12) The reduction in ductility is associated with an increased tendency to grain boundary fracture. In thermal reactors it is produced mainly by the formation of helium atoms by the reaction of thermal neutrons with the B¹⁰ isotope in the natural boron impurity in the alloys.⁽¹¹⁻¹³⁾

In a neutron spectrum which contains mainly fast neutrons some helium is generated by the B¹⁰ (*n,α*) Li⁷ reaction but considerable quantities arise from (*n,α*) reactions between very fast neutrons and the isotopes of many of the constituent elements in austenitic steels and nickel base alloys. Similarly (*n,p*) reactions also occur. However, hydrogen should quickly diffuse out of the alloys; therefore it is unlikely to contribute to the irradiation effects.

The helium produced by (*n,α*) reactions in isotopes other than B¹⁰ will be distributed homogeneously initially; that produced from B¹⁰ will, to some extent, be a function of the distribution of the boron and this depends upon the type and history of the material. However, the recoil distance for an α-particle produced by transmutation of B¹⁰ is about 2×10^{-4} cm, so that recoils will tend to disperse the helium from this source. Since the solubility of inert gases in metals is very low, the helium should precipitate to form bubbles at nucleation sites such as dislocation lines, grain boundaries and clusters of point defects where there are adequate supplies of vacancies.⁽¹⁴⁾ Electron microscope observations of the behaviour of helium bubbles in copper⁽¹⁵⁾ and theoretical considerations⁽¹⁶⁾ indicate that the bubbles migrate by diffusion of the matrix atoms over the surfaces of the bubbles. In the absence of a driving force they migrate randomly by Brownian motion and they are likely to grow by the collision and amalgamation of bubbles.

If a driving force such as a temperature, stress or concentration gradient is applied, the bubbles move in the direction of the applied force so that the energy of the system is lowered. Furthermore, if a bubble is attached to a dislocation line or a grain boundary which can lower the energy of the system by moving, there will again be a force acting on the bubble. Since the maximum force exerted on a bubble by a grain boundary is generally greater than that due to a dislocation, an applied stress is likely to sweep bubbles into grain boundaries.⁽¹⁶⁾ These helium bubbles at grain boundaries can cause intergranular embrittlement.

The gas pressure P_0 within a spherical bubble of radius r_0 at equilibrium on a grain boundary is given by

$$P_0 = 2\gamma/r_0 \quad (3)$$

where γ is the surface tension of the bubble. If a tensile stress σ is applied perpendicular to the grain boundary, the bubble is subjected to a hydrostatic stress and it grows to a new equilibrium radius with a reduced internal pressure. It has been shown^(17, 18) that, if the stress exceeds the critical value ($\sigma_c = 0.76\gamma/r_0$), equilibrium cannot be maintained and the bubble continues

to grow, at a rate determined by the temperature and the stress, by acquiring vacancies from the grain boundary. Grain boundary failure eventually occurs by coalescence of the bubbles. Evidence in support of this mechanism has been provided by electron microscope examination of specimens of irradiated austenitic steels which have been fractured at temperatures above 650°C. These show the presence of voids at grain boundaries. These voids cover the range from small individual spherical bubbles to cracks at grain boundaries perpendicular to the stress axis.

The deleterious effects of the helium bubbles at grain boundaries can be reduced by grain refinement, and thus increasing the grain boundary area. This can reduce the creep strength of the material prior to irradiation but it may lessen the loss of creep strength due to irradiation damage. This requires further investigation. Strengthening the grain boundaries by precipitation of stable carbides at the grain boundaries also reduces the loss of high temperature ductility due to helium bubble formation. We have found that, in thermal reactor irradiations, the FV548 material, which contains a fine dispersion of niobium carbide precipitate at the grain boundaries, is less susceptible to loss of high temperature ductility than 316 material. Reducing the grain size of FV548 material also makes it less susceptible to high temperature embrittlement after irradiation.

7. Conclusions

The information obtained on the rates of corrosion of austenitic and ferritic steels in sodium shows that, provided low oxygen concentrations are maintained in the sodium, serious corrosion problems should not arise in sodium cooled reactors at temperatures at least up to 650°C. If reasonable precautions are taken to reduce the amount of carbon in the sodium introduced into the system initially, and care is taken in using materials which do not decarburize, carburization of stainless steel should not present a serious problem.

Good progress has been made on the effect of neutron irradiation on the creep and yield properties and the ductility of potential cladding materials. More work is required before a final choice can be made.

References

1. H. LAWTON, K. Q. BAGLEY, E. EDMONDS and H. E. TILBE, The Irradiation Behaviour of Plutonium-bearing Ceramic Fuel pins, this Symposium, Paper 48/4.
2. J. W. EVANS and A. W. THORLEY, *Corrosion of Niobium and Vanadium in Liquid Sodium*, UKAEA Report IGR-TN/C1019 (1958).
3. A. W. THORLEY and C. TYZACK, in *Thermodynamics of Nuclear Materials*, IAEA, Vienna, pp. 365-97 (1962).
4. R. W. LOCKHART and G. BILLURIS, in *Nuclear Metallurgy*, Vol. IX, AIMME, New York, pp. 207-25 (1963).

5. R. HARDE and K. W. STÖHR, *Third U.N. Int. Conf. on the Peaceful Uses of Atomic Energy*, A/CONF. 28/P/537 (1964).
6. D. H. GURINSKY, J. R. WEEKS, C. J. KLAMUT, L. ROSENBLUM and J. H. DEVAN, *Third U.N. Int. Conf. on the Peaceful Uses of Atomic Energy*, A/CONF. 28/P/244 (1964).
7. B. A. NEVSOROV and V. V. ZOTOV, *Third U.N. Int. Conf. on the Peaceful Uses of Atomic Energy*, A/CONF. 28/P/343 (1964).
8. G. ILINCEV, *Jaderna Energie* **10**, 2 (1964).
9. J. F. W. BISHOP, C. BOORMAN, G. O. JACKSON, D. L. LINNING, N. PARKINSON and A. E. WILLIAMS, Design and Development of Fuel and Fuel Elements for Prototype Fast Reactor, this Symposium, Paper 4B/1.
10. R. S. BARNES, A.S.T.M. Special Technical Publication No. 380, 40 (1965).
11. P. C. L. PFEIL and D. R. HARRIES, *ibid.* 202.
12. D. R. HARRIES, K. Q. BAGLEY, I. P. BELL, W. S. GIBSON, J. GILLIES, P. C. L. PFEIL and S. B. WRIGHT, *Third U.N. Int. Conf. on the Peaceful Uses of Atomic Energy*, A/CONF. 28/P/162 (1964).
13. P. R. B. HIGGINS and A. C. ROBERTS, *Nature* **206**, 1249 (1965).
14. R. S. BARNES and D. J. MAZEY, *Phil. Mag.* **5**, 1247 (1960).
15. R. S. BARNES and D. J. MAZEY, *Proc. Roy. Soc. A* **275**, 47 (1963).
16. R. S. BARNES and R. S. NELSON, AERE Report R4952 (1965).
17. E. D. HYAM and G. SUMNER, in *Radiation Damage in Solids*, Vol. I, IAEA, Vienna, pp. 323–32 (1962).
18. R. S. BARNES, *Nature* **206**, 1307 (1965).